CSE490 (D)

Introduction to Football Data & Analytics

Research Project

[Team no 14]

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THE DEFENSIVE DISRUPTOR SCORE

Valuing Defensive Actions in Football
Using Event Data

THE PROBLEM

Current defensive statistics are TOO SIMPLE

Team	Tournament	Shots pg	Tackles pg	Interceptions pg	Fouls pg	Offsides pg	Rating
1. Tottenham	Premier League	12.9	19.7	7.7	11.6	1.6	6.58
2. Wolves	Premier League	14	19.4	8.6	13.3	1.3	6.47
3. Lens	Ligue 1	11.3	19.2	11.1	13.4	2.1	6.81
4. Brest	Ligue 1	13.5	19	10	13.4	1.1	6.64
5. Crystal Palace	Premier League	11.6	18.9	8.6	10.7	1.9	6.66
6. Deportivo Alaves	LaLiga	12.8	18.9	8	15.5	2	6.52
7. Getafe	LaLiga	10.7	18.6	8.3	15	2.4	6.54

- **ALL TACKLES COUNT THE SAME**
  - A tackle in your own box = same as midfield

- **NO CONTEXT ABOUT OUTCOME**
  - Did the tackle lead to possession? Or did opponent get ball back?

- **DEFENSIVE PLAYERS ARE UNDERVALUED**
  - Goal scorers get all the attention

MY SOLUTION: Create a "Disruptor Score" that gives context to defensive actions

DATASET: StatsBomb Open Data

DATASET LINK: https://github.com/statsbomb/open-data

DATA STRUCTURE:

- ✓ Competition (e.g., FIFA World Cup)
- ✓ Season (e.g., 2018, 2022)
- ✓ Match (e.g., England vs Colombia)
- ✓ EVENTS (each row = one action)

EVENT TYPES I'LL USE:

- ✓ 🛑 Tackles
- ✓ ✋ Interceptions
- ✓ 🧹 Clearances
- ✓ 😤 Pressures
- ✓ ⚽ Shots/Goals (to track outcomes)

Model/Method I Will You Use..

STEP 1: DATA FILTERING

- ❑ Select Premier League or LaLiga or 2018 World Cup
- ❑ Pick 10-15 defensive players
- ❑ Extract all defensive events

STEP 2: LOCATION WEIGHTING

- ❑ Divide pitch into 4 danger zones
- ❑ Assign points based on danger:
  - ✓ Own box = 5 points
  - ✓ Defensive third = 3 points
  - ✓ Midfield = 2 points
  - ✓ Opponent half = 1 point

STEP 3: OUTCOME ANALYSIS

- ❑ Look at next 3 events after each defensive action
- ❑ Classify outcomes:
  - ✓ Team kept possession = +2
  - ✓ Led to counter-attack = +3
  - ✓ Led to goal = +5
  - ✓ Opponent kept ball = -1

STEP 4: ADVANCED ANALYSIS

A. **PLAYER FINGERPRINTS**

- Radar charts for each player
- Shows defensive specialty (tackle expert? interceptor?)
- Cluster similar players together

B. **CONSISTENCY SCORING**

- Split season into 3 parts
- Calculate variance in performance
- Create "Reliability Rating" = Score × (1 - Variance)

C. **VISUALIZATION**

- Radar charts for top 5 players
- Consistency line charts
- Final ranking with reliability adjustment

FORMULA

Disruptor Score = Location Points + Outcome Bonus - Outcome Penalty

Final Reliable Score = Base Score × (1 - Consistency Penalty)

VALIDATION:

- ❑Compare to team goals conceded
- ❑Check if scores match expert opinion
- ❑Test consistency across season

REFERENCE PAPERS

1. Merhej, C., Beal, R. J., Matthews, T., & Ramchurn, S. (2021). What happened next? Using deep learning to value defensive actions in football Event-Data. Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD ’21)., 3394–3403. https://doi.org/10.1145/3447548.3467090
2. Kim, H., Seo, S., Choi, H., Boomstra, T., Yoon, J., & Park, C. (2025, December 11). Better Prevent than Tackle: Valuing Defense in Soccer Based on Graph Neural Networks. arXiv.org. https://arxiv.org/abs/2512.10355
3. Herold, M., Hecksteden, A., Radke, D., Goes, F., Nopp, S., Meyer, T., & Kempe, M. (2022). Off-ball behavior in association football: A data-driven model to measure changes in individual defensive pressure. Journal of Sports Sciences, 40(12), 1412–1425. https://doi.org/10.1080/02640414.2022.2081405